Project SENTINEL: Mechanistically Interpretable Steering of Foundation Models for De Novo Protein Binder Design

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Motivation

Engineered protein biosensors offer real-time disease detection, but the current practice to develop these biosensors requires extensive human effort to design and validate them. Protein language models (PLMs) are an evolving methodology that alleviates the manual overhead involved in the design process, but generative binder-design models remain black boxes. In their current state, PLMs produce candidate binders without characterizing the internal reasoning that separates a high-affinity binder from a poor one. This opacity limits rational design improvement, failure-mode diagnosis, and cross-disease adaptation.

Approach

We train sparse autoencoders (SAEs) on ESM-C PLM's residue-level activations that we extracted from Boltz-designed binder candidates, to recover human-readable features present in the PLM's underlying latent representation. As proof-of-concept, we validate this pipeline through an end-to-end evaluation on Vilip-1, a blood-circulating biomarker of acute neurological injury.

Results

- Diversifying SAE training data: folding in real, evolutionarily distinct sequences alongside synthetic designs, and mixing in binder-design data from additional targets (UCH-L1, B-FABP) substantially improved generalization to real proteins beyond synthetic designs alone.
- Independently trained SAE dictionaries: including one incorporating binder-target co-attention context via ESM-C's native chain-break token converges on the same residues in real proteins for a subset of examined features, corroborated by InterPro domain annotations and calibrated LLM-assisted labeling.
- This convergence holds up against the confound of shared random initialization across runs: the agreeing dictionaries mostly rely on different learned feature indices to detect the same real residues, evidence of genuine independent convergence rather than an artifact of training setup.

Ongoing & Future Work

This interpretability evaluation is a necessary precursor to our broader goal of steering generative design toward high-affinity binding features, analogous to concept-guided steering in text-to-image diffusion models, rather than blindly sampling the latent space. We plan to extend this pipeline to three clinical targets (UCH-L1, S100B, and B-FABP) and evaluate whether SAE features predict binding-quality metrics (binding confidence, ipTM, ipSAE) directly, thus helping establish the value of mechanistic interpretability in accelerating PLM-informed biosensor design.